

Nombre: KEVIN SAMUEL HILASACA BARRIOS

COLUMNA CIRCULAR

Elaborado por Kevin HB

PRÁCTICA CALIFICADA - EJERCICIO INDIVIDUAL

Construir el diagrama de interacción uniaxial en el formato de su preferencia (Excel, Mathcad, Smarth Solver) con respecto al eje "x" y "y" de la columna asignada, considere concreto $f'_c = 210 \text{ kg/cm}^2$, acero de refuerzo $f_y = 4200 \text{ kg/cm}^2$, recubrimiento neto al estribo 5 cm., explicando paso a paso la interacción entre su configuración estructural.

Materiales:

$F_c = 210 \text{ kg/cm}^2$; Resistencia a la compresión del concreto

$F_y = 4200 \text{ kg/cm}^2$; Fluencia del acero

Módulos de Elasticidad:

$E_c = 217370.7 \text{ kg/cm}^2$; Elasticidad del concreto $15000 \times \sqrt{F_c}$

$E_s = 2000000 \text{ kg/cm}^2$; Elasticidad del acero 2×10^6

Deformaciones:

$\epsilon_c = 0.003$; Deformación Unitaria del concreto a compresión

$\epsilon_s = 0.0021$; Deformación Unitaria del Acero a Fluencia

Geometría:

$r = 5.00 \text{ cm}$; Recubrimiento neto al estribo

$\beta_1 = 0.85$; Factor adimensional

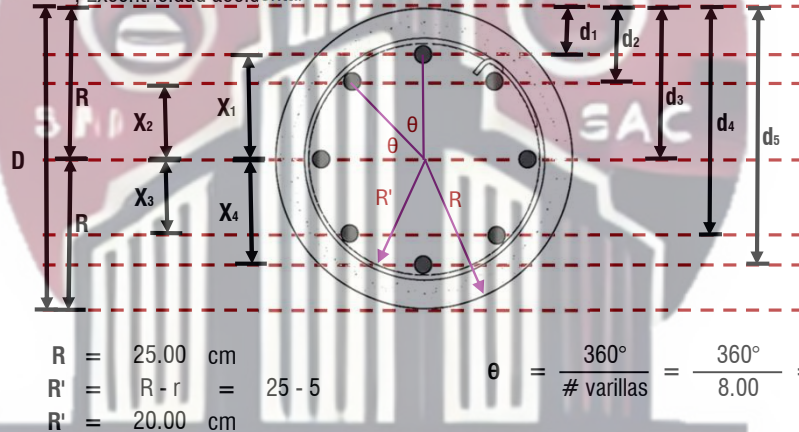
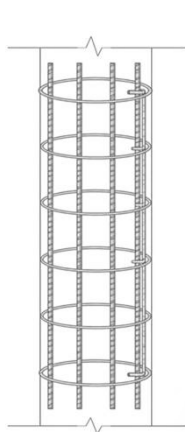
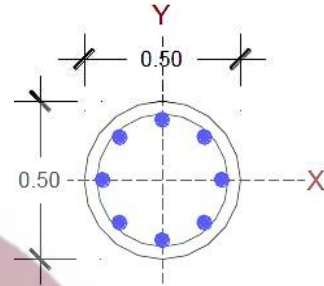
$\Phi = 0.70$ estribado ; Coeficiente de reducción E.060

$\alpha = 0.80$; Excentricidad accidental

$D = 50.00 \text{ cm}$; Diámetro

$R = 25.00 \text{ cm}$; Radio

$8 \text{ } \phi \text{ } 5/8"$; varillas



[A] CÁLCULO DE PERALTES

$X_1 = R' = 20 \text{ cm}$	$= 20 \text{ cm}$	$d_1 = R - X_1 = 25 - 20 = 5.00 \text{ cm}$
$X_2 = R' \cos \theta = 20 \cos(45)$	$= 14.1 \text{ cm}$	$d_2 = 25 - 14.14 = 10.9 \text{ cm}$
$X_3 = 0$	$= 0.0 \text{ cm}$	$d_3 = 25 - 0 = 25.0 \text{ cm}$
$X_4 = R' \cos \theta = 20 \cos(45)$	$= 14.1 \text{ cm}$	$d_4 = 25 - (-14.14) = 39.1 \text{ cm}$
$X_5 = R' = 20 \text{ cm}$	$= 20 \text{ cm}$	$d_5 = 25 - (-20) = 45.0 \text{ cm}$

[B] ÁREA DE ZONA COMPRIMIDA DE COLUMNA (A_c)

$$A_c = \left[\frac{\theta \pi}{180} - \sin \theta \cos \theta \right] * R^2$$

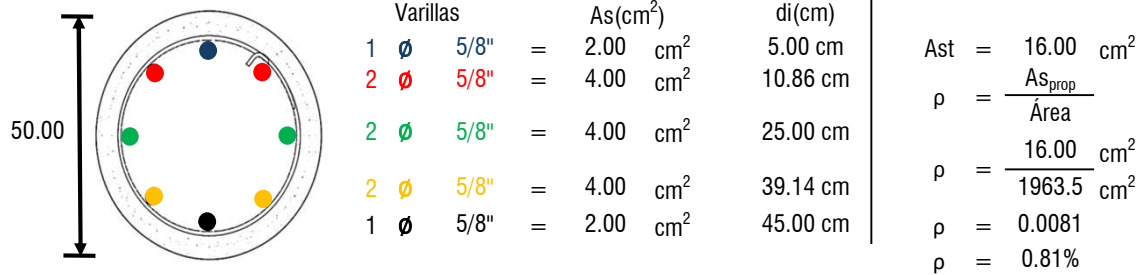
$$A_c = K_1 * R^2$$

Altura del bloque equivalente de refuerzo (a)

$$a = \beta_1 * C$$

$$R - a = R \cos \theta \quad \longrightarrow \quad \theta = \cos^{-1} \left(\frac{R - a}{R} \right)$$

[C] DIAGRAMACIÓN

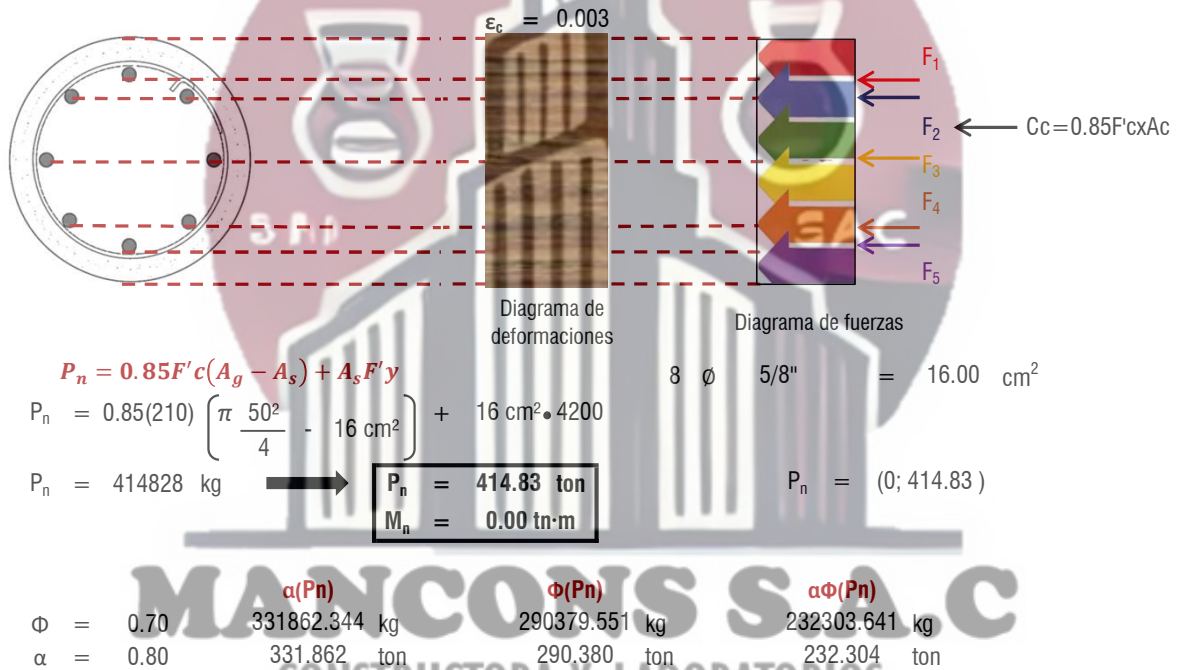


[D] CENTROIDE PLÁSTICO

$$C_p = \frac{(Y_{cg})(0.85 \cdot F'_c \cdot A_g) + (\sum A_{si} \cdot d_i)(F'_y)}{0.85(F'_c)A_g + F'_y \sum A_{si}} = \frac{(25\text{cm})(0.85 \times 210 \text{ kg/cm}^2 \times 1963.5) + (16 \times A_{si} \cdot d_i)(4200)}{0.85(210)1963.5 + 4200(A_{si})}$$

$$C_p = 25.000 \text{ cm}$$

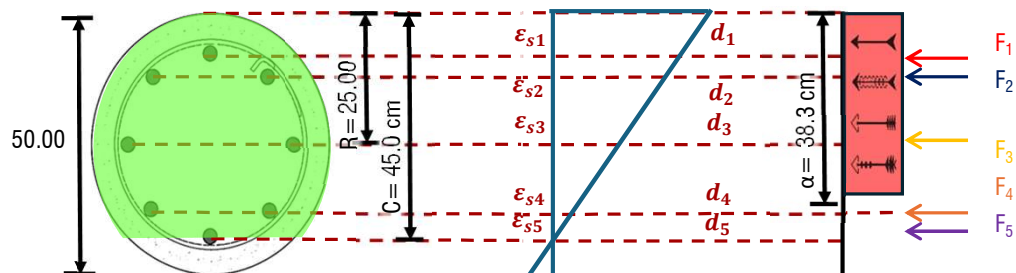
[1] COMPRESIÓN PURA



[2] FISURACIÓN INCIPIENTE

$$\alpha = \beta_1 C$$

$c = 45.0 \text{ cm}$
 $\alpha = 38.3 \text{ cm}$
 $E_s = 2000000 \text{ kg/cm}^2$
 $D = 50.0 \text{ cm}$
 $\epsilon_c = 0.003$
 $F'_y = 4200 \text{ kg/cm}^2$
 $F'_c = 210 \text{ kg/cm}^2$



Deformaciones de cada acero

$$\epsilon_{si} = \left(\frac{C - d_i}{C} \right) * \epsilon_c$$

Fuerzas del acero F_y $\epsilon_s = 0.0021 = \epsilon_y$

$$f_{si} = \epsilon_{si} * E_s$$

$$\epsilon_{s1} = \left(\frac{C - d_1}{C} \right) * \epsilon_c \rightarrow \epsilon_{s1} = 0.00267 \quad \text{As FLUYE} \quad f_{s1} = \epsilon_{s1} * E_s \rightarrow f_{s1} = 5333.3 \text{ kg/cm}^2 \rightarrow 4200 \text{ kg/cm}^2$$

$$\epsilon_{s2} = \left(\frac{C - d_2}{C} \right) * \epsilon_c \rightarrow \epsilon_{s2} = 0.00228 \quad \text{As FLUYE} \quad f_{s2} = \epsilon_{s2} * E_s \rightarrow f_{s2} = 4552.3 \text{ kg/cm}^2 \rightarrow 4200 \text{ kg/cm}^2$$

$$\epsilon_{s3} = \left(\frac{C - d_3}{C} \right) * \epsilon_c \rightarrow \epsilon_{s3} = 0.00133 \quad \text{As NO FLUYE} \quad f_{s3} = \epsilon_{s3} * E_s \rightarrow f_{s3} = 2666.7 \text{ kg/cm}^2 \rightarrow 2666.7 \text{ kg/cm}^2$$

$$\epsilon_{s4} = \left(\frac{C - d_4}{C} \right) * \epsilon_c \rightarrow \epsilon_{s4} = 0.00039 \quad \text{As NO FLUYE} \quad f_{s4} = \epsilon_{s4} * E_s \rightarrow f_{s4} = 781.05 \text{ kg/cm}^2 \rightarrow 781.05 \text{ kg/cm}^2$$

$$\epsilon_{s5} = \left(\frac{C - d_5}{C} \right) * \epsilon_c \rightarrow \epsilon_{s5} = 0.00000 \quad \text{As NO FLUYE} \quad f_{s5} = \epsilon_{s5} * E_s \rightarrow f_{s5} = 0 \text{ kg/cm}^2 \rightarrow 0 \text{ kg/cm}^2$$

Fuerzas de cada capa de acero

$$F_{si} = f_{si} * A_{si}$$

$$F_{s1} = f_{s1} * A_{s1} \rightarrow 8400.0 \text{ kg} \rightarrow C_{s1} = 8.40 \text{ Ton} \quad F_{s4} = f_{s4} * A_{s4} \rightarrow 3124.2 \text{ kg} \rightarrow C_{s4} = 3.12 \text{ Ton}$$

$$F_{s2} = f_{s2} * A_{s2} \rightarrow 16800.0 \text{ kg} \rightarrow C_{s2} = 16.80 \text{ Ton} \quad F_{s5} = f_{s5} * A_{s5} \rightarrow 0.0 \text{ kg} \rightarrow T_{s5} = 0.00 \text{ Ton}$$

$$F_{s3} = f_{s3} * A_{s3} \rightarrow 10666.7 \text{ kg} \rightarrow C_{s3} = 10.67 \text{ Ton}$$

Cálculo de Ac

$$\theta = \cos^{-1} \left(\frac{R - a}{R} \right) \rightarrow \arccos \left(\frac{25 - 38.25}{25.00} \right) = 122.01^\circ \quad A_c = \left[\frac{\theta \pi}{180^\circ} - \sin(\theta) \cos(\theta) \right] R^2 = 1611.77 \text{ cm}^2$$

Cálculo de Cc

$$C_{cc} = 0.85 * F'_c(A_c) \rightarrow C_{cc} = 287701 \text{ kg} = 287.70 \text{ Ton}$$

Fuerzas Resultantes

$$P_n = C_{s1} + C_{s2} + C_{s3} + C_{s4} - T_{s5} + C_{cc} = 326.69 \text{ Ton}$$

$$M_n = C_{s1} \left(\frac{h}{2} - d_1 \right) + C_{s2} \left(\frac{h}{2} - d_2 \right) + C_{s3} \left(d_3 - \frac{h}{2} \right) + C_{s4} \left(d_4 - \frac{h}{2} \right) + T_{s5} \left(d_5 - \frac{h}{2} \right) + C_{cc} \left(\frac{h}{2} - \frac{a}{2} \right) =$$

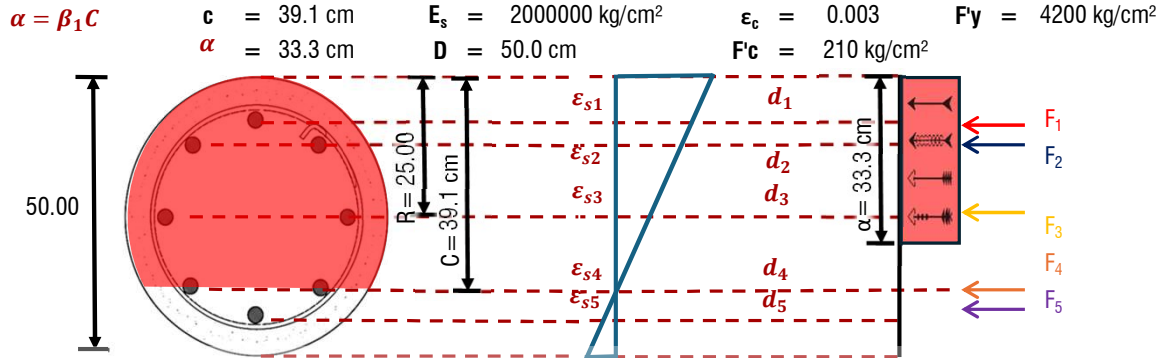
Fuerza (ton)	Brazo (m)	Momento (tn·m)
$C_{s1} = 8.40 \text{ Ton}$	0.20 m	1.68 tn·m
$C_{s2} = 16.8 \text{ Ton}$	0.14 m	2.38 tn·m
$C_{s3} = 10.7 \text{ Ton}$	0.00 m	0.00 tn·m
$C_{s4} = 3.1 \text{ Ton}$	0.14 m	0.44 tn·m
$T_{s5} = 0.00 \text{ Ton}$	0.20 m	0.00 tn·m
$C_{cc} = 288 \text{ Ton}$	0.06 m	16.90 tn·m
Mn=		21.40 tn·m
Pn=		326.69 Ton



Varillas	As (cm²)	di (cm)
1 Ø 5/8" =	2.00 cm²	5.00 cm
2 Ø 5/8" =	4.00 cm²	10.86 cm
2 Ø 5/8" =	4.00 cm²	25.00 cm
2 Ø 5/8" =	4.00 cm²	39.14 cm
1 Ø 5/8" =	2.00 cm²	45.00 cm

[3] ZONA FALLA FRAGIL

C > C_b



Deformaciones de cada acero

Fuerzas del acero F_y $\epsilon_s = 0.0021 = \epsilon_y$

$$\epsilon_{si} = \left(\frac{c - d_i}{c} \right) * \epsilon_c$$

$$f_{si} = \epsilon_{si} * E_s$$

$\epsilon_{s1} = \left(\frac{c - d_1}{c} \right) * \epsilon_c \rightarrow \epsilon_{s1} = 0.00262$	As FLUYE	$f_{s1} = \epsilon_{s1} * E_s \rightarrow f_{s1} = 5233.6 \text{ kg/cm}^2 \rightarrow 4200 \text{ kg/cm}^2$
$\epsilon_{s2} = \left(\frac{c - d_2}{c} \right) * \epsilon_c \rightarrow \epsilon_{s2} = 0.00217$	As FLUYE	$f_{s2} = \epsilon_{s2} * E_s \rightarrow f_{s2} = 4335.6 \text{ kg/cm}^2 \rightarrow 4200 \text{ kg/cm}^2$
$\epsilon_{s3} = \left(\frac{c - d_3}{c} \right) * \epsilon_c \rightarrow \epsilon_{s3} = 0.00108$	As NO FLUYE	$f_{s3} = \epsilon_{s3} * E_s \rightarrow f_{s3} = 2167.8 \text{ kg/cm}^2 \rightarrow 2167.8 \text{ kg/cm}^2$
$\epsilon_{s4} = \left(\frac{c - d_4}{c} \right) * \epsilon_c \rightarrow \epsilon_{s4} = 0.00000$	As NO FLUYE	$f_{s4} = \epsilon_{s4} * E_s \rightarrow f_{s4} = 0 \text{ kg/cm}^2 \rightarrow 0 \text{ kg/cm}^2$
$\epsilon_{s5} = \left(\frac{c - d_5}{c} \right) * \epsilon_c \rightarrow \epsilon_{s5} = 0.0011$	As NO FLUYE	$f_{s5} = \epsilon_{s5} * E_s \rightarrow f_{s5} = 2115.2 \text{ kg/cm}^2 \rightarrow 2115.2 \text{ kg/cm}^2$

Fuerzas de cada capa de acero

$$F_{si} = f_{si} * A_{si}$$

$F_{s1} = f_{s1} * A_{s1} \rightarrow 8400.0 \text{ kg} \rightarrow C_{s1} = 8.40 \text{ Ton}$	$F_{s4} = f_{s4} * A_{s4} \rightarrow 0.0 \text{ kg} \rightarrow T_{s4} = 0.00 \text{ Ton}$
$F_{s2} = f_{s2} * A_{s2} \rightarrow 16800.0 \text{ kg} \rightarrow C_{s2} = 16.80 \text{ Ton}$	$F_{s5} = f_{s5} * A_{s5} \rightarrow 4230.4 \text{ kg} \rightarrow T_{s5} = 4.23 \text{ Ton}$
$F_{s3} = f_{s3} * A_{s3} \rightarrow 8671.3 \text{ kg} \rightarrow C_{s3} = 8.67 \text{ Ton}$	

Cálculo de A_c

$$\theta = \cos^{-1} \left(\frac{R - \alpha}{R} \right) \rightarrow \arccos \left(\frac{25 - 33.27}{25.00} \right) = 109.32^\circ$$

$$A_c = \left[\frac{\theta \pi}{180^\circ} - \sin(\theta) \cos(\theta) \right] R^2 = 1387.62 \text{ cm}^2$$

Cálculo de C_c

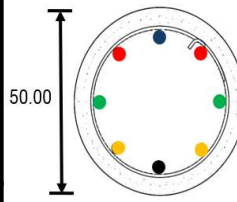
$$C_{cc} = 0.85 * F'_c(A_c) \rightarrow C_{cc} = 247689 \text{ kg} = 247.69 \text{ Ton}$$

Fuerzas Resultantes

$$P_n = C_{s1} + C_{s2} + C_{s3} - T_{s4} - T_{s5} + C_{cc} = 277.33 \text{ Ton}$$

$$M_n = C_{s1} \left(\frac{h}{2} - d_1 \right) + C_{s2} \left(\frac{h}{2} - d_2 \right) + C_{s3} \left(d_3 - \frac{h}{2} \right) + T_{s4} \left(d_4 - \frac{h}{2} \right) + T_{s5} \left(d_5 - \frac{h}{2} \right) + C_{cc} \left(\frac{h}{2} - \frac{a}{2} \right) =$$

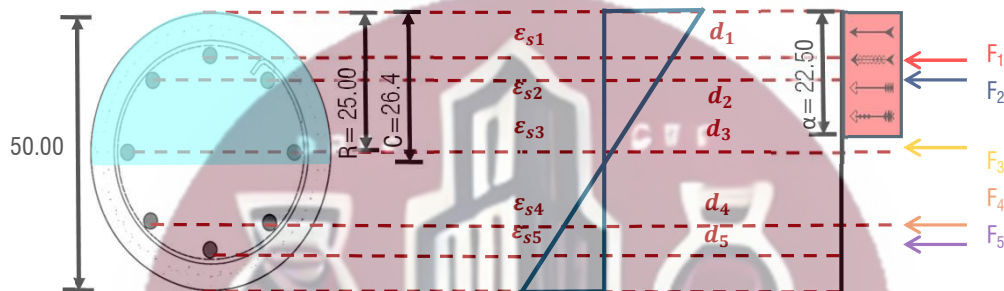
Fuerza (ton)	Brazo (m)	Momento (tn·m)
$C_{s1} = 8.40$ Ton	0.20 m	1.68 tn·m
$C_{s2} = 16.8$ Ton	0.14 m	2.38 tn·m
$C_{s3} = 8.7$ Ton	0.00 m	0.00 tn·m
$T_{s4} = 0.0$ Ton	0.14 m	0.00 tn·m
$T_{s5} = 4.23$ Ton	0.20 m	0.85 tn·m
$C_{cc} = 248$ Ton	0.08 m	20.72 tn·m
Mn =		25.62 tn·m
Pn =		277.33 Ton



Varillas		As(cm ²)	di(cm)
1 Ø 5/8"	=	2.00 cm ²	5.00 cm
2 Ø 5/8"	=	4.00 cm ²	10.86 cm
2 Ø 5/8"	=	4.00 cm ²	25.00 cm
2 Ø 5/8"	=	4.00 cm ²	39.14 cm
1 Ø 5/8"	=	2.00 cm ²	45.00 cm

[4] CARGA BALANCEADA

$$C_b = \left(\frac{6000}{6000 + F'y} \right) * d_4 = \left(\frac{6000}{6000 + 4200} \right) * 45 = 26.471 \text{ cm} \quad \left| \begin{array}{l} \alpha = \beta_1 C = 0.85 * 26.47 \\ \alpha = 22.50 \text{ cm} \end{array} \right.$$



Deformaciones de cada acero

$$\epsilon_{si} = \left(\frac{C - d_i}{C} \right) * \epsilon_c$$

Fuerzas del acero F'y $\epsilon_s = 0.0021 = \epsilon_y$

$$f_{si} = \epsilon_{si} * E_s$$

$\epsilon_{s1} = \left(\frac{C - d_1}{C} \right) * \epsilon_c \rightarrow \epsilon_{s1} = 0.00243$	As FLUYE	$f_{s1} = \epsilon_{s1} * E_s \rightarrow f_{s1} = 4866.7 \text{ kg/cm}^2 \rightarrow 4200 \text{ kg/cm}^2$
$\epsilon_{s2} = \left(\frac{C - d_2}{C} \right) * \epsilon_c \rightarrow \epsilon_{s2} = 0.00177$	As NO FLUYE	$f_{s2} = \epsilon_{s2} * E_s \rightarrow f_{s2} = 3538.9 \text{ kg/cm}^2 \rightarrow 3538.9 \text{ kg/cm}^2$
$\epsilon_{s3} = \left(\frac{C - d_3}{C} \right) * \epsilon_c \rightarrow \epsilon_{s3} = 0.00017$	As NO FLUYE	$f_{s3} = \epsilon_{s3} * E_s \rightarrow f_{s3} = 333.33 \text{ kg/cm}^2 \rightarrow 333.33 \text{ kg/cm}^2$
$\epsilon_{s4} = \left(\frac{C - d_4}{C} \right) * \epsilon_c \rightarrow \epsilon_{s4} = 0.0014$	As NO FLUYE	$f_{s4} = \epsilon_{s4} * E_s \rightarrow f_{s4} = 2872.2 \text{ kg/cm}^2 \rightarrow 2872.2 \text{ kg/cm}^2$
$\epsilon_{s5} = \left(\frac{C - d_5}{C} \right) * \epsilon_c \rightarrow \epsilon_{s5} = 0.0021$	As FLUYE	$f_{s5} = \epsilon_{s5} * E_s \rightarrow f_{s5} = 4200 \text{ kg/cm}^2 \rightarrow 4200 \text{ kg/cm}^2$

Fuerzas de cada capa de acero

$$F_{si} = f_{si} * A_{si}$$

$$\begin{aligned} F_{s1} &= f_{s1} * A_{s1} \rightarrow 8400.0 \text{ kg} \rightarrow C_{s1} = 8.40 \text{ Ton} & F_{s4} &= f_{s4} * A_{s4} \rightarrow 11488.9 \text{ kg} \rightarrow T_{s4} = 11.49 \text{ Ton} \\ F_{s2} &= f_{s2} * A_{s2} \rightarrow 14155.5 \text{ kg} \rightarrow C_{s2} = 14.16 \text{ Ton} & F_{s5} &= f_{s5} * A_{s5} \rightarrow 8400.0 \text{ kg} \rightarrow T_{s5} = 8.40 \text{ Ton} \\ F_{s3} &= f_{s3} * A_{s3} \rightarrow 1333.3 \text{ kg} \rightarrow T_{s3} = 1.33 \text{ Ton} \end{aligned}$$

Cálculo de A_c

$$\theta = \cos^{-1}\left(\frac{R-a}{R}\right) \rightarrow \arccos\left(\frac{25-22.5}{25.00}\right) = 84.26^\circ \quad A_c = \left[\frac{\theta\pi}{180^\circ} - \sin(\theta)\cos(\theta)\right] R^2 = 856.96 \text{ cm}^2$$

Cálculo de C_c

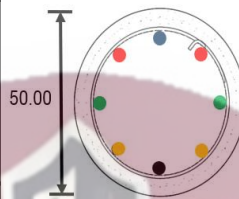
$$C_{cc} = 0.85 * F'_c(A_c) \rightarrow C_{cc} = 152967 \text{ kg} = 152.97 \text{ Ton}$$

Fuerzas Resultantes

$$P_n = C_{s1} + C_{s2} - T_{s3} - T_{s4} + T_{s5} + C_{cc} = 154.30 \text{ Ton}$$

$$M_n = C_{s1}\left(\frac{h}{2} - d_1\right) + C_{s2}\left(\frac{h}{2} - d_2\right) + T_{s3}\left(d_3 - \frac{h}{2}\right) - T_{s4}\left(d_4 - \frac{h}{2}\right) + T_{s5}\left(d_5 - \frac{h}{2}\right) + C_{cc}\left(\frac{h}{2} - \frac{a}{2}\right) =$$

Fuerza (ton)	Brazo (m)	Momento (tn·m)
$C_{s1} = 8.40 \text{ Ton}$	0.20 m	1.68 tn·m
$C_{s2} = 14.2 \text{ Ton}$	0.14 m	2.00 tn·m
$T_{s3} = 1.33 \text{ Ton}$	0.00 m	0.00 tn·m
$T_{s4} = 11.5 \text{ Ton}$	0.14 m	1.62 tn·m
$T_{s5} = 8.40 \text{ Ton}$	0.20 m	1.68 tn·m
$C_{cc} = 153 \text{ Ton}$	0.14 m	21.03 tn·m
Mn=		28.02 tn·m
Pn=		154.30 Ton



Varillas	As(cm ²)	di(cm)
1 Ø 5/8" =	2.00 cm ²	5.00 cm
2 Ø 5/8" =	4.00 cm ²	10.86 cm
2 Ø 5/8" =	4.00 cm ²	25.00 cm
2 Ø 5/8" =	4.00 cm ²	39.14 cm
1 Ø 5/8" =	2.00 cm ²	45.00 cm

[5] ZONA DE FALLA DUCTIL

$C < C_b$

$$\alpha = \beta_1 C$$

$$c = 10.9 \text{ cm}$$

$$E_s = 2000000 \text{ kg/cm}^2$$

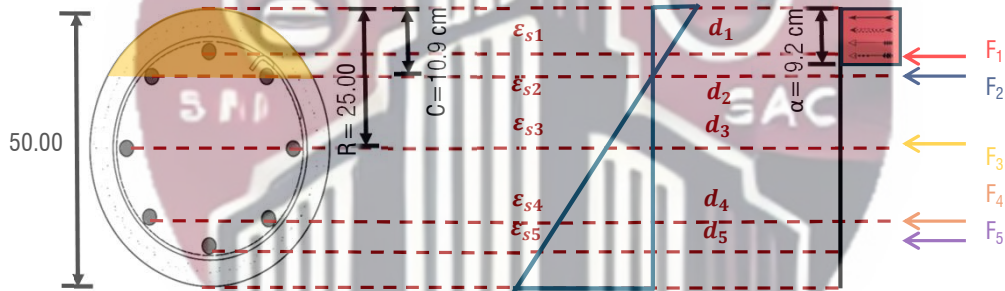
$$\epsilon_c = 0.003$$

$$F_y = 4200 \text{ kg/cm}^2$$

$$\alpha = 9.2 \text{ cm}$$

$$D = 50.0 \text{ cm}$$

$$F'_c = 210 \text{ kg/cm}^2$$



Deformaciones de cada acero

$$\text{Fuerzas del acero } F_y \quad \epsilon_s = 0.0021 = \epsilon_y$$

$$\epsilon_{si} = \left(\frac{C-d_i}{C}\right) * \epsilon_c$$

$$f_{si} = \epsilon_{si} * E_s$$

$$\epsilon_{s1} = \left(\frac{C-d_1}{C}\right) * \epsilon_c \rightarrow \epsilon_{s1} = 0.00162 \quad \text{As NO FLUYE} \quad f_{s1} = \epsilon_{s1} * E_s \rightarrow f_{s1} = 3237 \text{ kg/cm}^2 \rightarrow 3237 \text{ kg/cm}^2$$

$$\epsilon_{s2} = \left(\frac{C-d_2}{C}\right) * \epsilon_c \rightarrow \epsilon_{s2} = 0.00000 \quad \text{As NO FLUYE} \quad f_{s2} = \epsilon_{s2} * E_s \rightarrow f_{s2} = 0 \text{ kg/cm}^2 \rightarrow 0 \text{ kg/cm}^2$$

$$\epsilon_{s3} = \left(\frac{C-d_3}{C}\right) * \epsilon_c \rightarrow \epsilon_{s3} = 0.0039 \quad \text{As FLUYE} \quad f_{s3} = \epsilon_{s3} * E_s \rightarrow f_{s3} = 7814.9 \text{ kg/cm}^2 \rightarrow 4200 \text{ kg/cm}^2$$

$$\epsilon_{s4} = \left(\frac{C-d_4}{C}\right) * \epsilon_c \rightarrow \epsilon_{s4} = 0.0078 \quad \text{As FLUYE} \quad f_{s4} = \epsilon_{s4} * E_s \rightarrow f_{s4} = 15630 \text{ kg/cm}^2 \rightarrow 4200 \text{ kg/cm}^2$$

$$\epsilon_{s5} = \left(\frac{C-d_5}{C}\right) * \epsilon_c \rightarrow \epsilon_{s5} = 0.0116 \quad \text{As FLUYE} \quad f_{s5} = \epsilon_{s5} * E_s \rightarrow f_{s5} = 23255 \text{ kg/cm}^2 \rightarrow 4200 \text{ kg/cm}^2$$

Fuerzas de cada capa de acero

$$F_{si} = f_{si} \cdot A_{si}$$

$$F_{s1} = f_{s1} \cdot A_{s1} \rightarrow 6474.1 \text{ kg} \rightarrow C_{s1} = 6.47 \text{ Ton} \quad F_{s4} = f_{s4} \cdot A_{s4} \rightarrow 16800.0 \text{ kg} \rightarrow T_{s4} = 16.80 \text{ Ton}$$

$$F_{s2} = f_{s2} \cdot A_{s2} \rightarrow 0.0 \text{ kg} \rightarrow T_{s2} = 0.00 \text{ Ton} \quad F_{s5} = f_{s5} \cdot A_{s5} \rightarrow 8400.0 \text{ kg} \rightarrow T_{s5} = 8.40 \text{ Ton}$$

$$F_{s3} = f_{s3} \cdot A_{s3} \rightarrow 16800.0 \text{ kg} \rightarrow T_{s3} = 16.80 \text{ Ton}$$

Cálculo de Ac

$$\theta = \cos^{-1} \left(\frac{R-a}{R} \right) \rightarrow \arccos \left(\frac{25-9.23}{25.00} \right) = 50.89^\circ \quad A_c = \left[\frac{\theta \pi}{180^\circ} - \sin(\theta) \cos(\theta) \right] R^2 = 249.18 \text{ cm}^2$$

Cálculo de Cc

$$C_{cc} = 0.85 \cdot F'_c(A_c) \rightarrow C_{cc} = 44479 \text{ kg} = 44.48 \text{ Ton}$$

Fuerzas Resultantes

$$P_n = C_{s1} - T_{s2} - T_{s3} - T_{s4} - T_{s5} + C_{cc} = 42.55 \text{ Ton}$$

$$M_n = C_{s1} \left(\frac{h}{2} - d_1 \right) + C_{s2} \left(\frac{h}{2} - d_2 \right) + C_{s3} \left(d_3 - \frac{h}{2} \right) + T_{s4} \left(d_4 - \frac{h}{2} \right) + T_{s5} \left(d_5 - \frac{h}{2} \right) + C_{cc} \left(\frac{h}{2} - \frac{a}{2} \right) =$$

Fuerza (ton)	Brazo (m)	Momento (tn·m)
$C_{s1} = 6.47 \text{ Ton}$	0.20 m	1.29 tn·m
$C_{s2} = 0.0 \text{ Ton}$	0.14 m	0.00 tn·m
$C_{s3} = 16.8 \text{ Ton}$	0.00 m	0.00 tn·m
$T_{s4} = 16.8 \text{ Ton}$	0.14 m	2.38 tn·m
$T_{s5} = 8.40 \text{ Ton}$	0.20 m	1.68 tn·m
$C_{cc} = 44 \text{ Ton}$	0.20 m	9.07 tn·m
Mn =		14.42 tn·m
Pn =		42.55 Ton



Varillas	As (cm ²)	di (cm)
1 Ø 5/8" =	2.00 cm ²	5.00 cm
2 Ø 5/8" =	4.00 cm ²	10.86 cm
2 Ø 5/8" =	4.00 cm ²	25.00 cm
2 Ø 5/8" =	4.00 cm ²	39.14 cm
1 Ø 5/8" =	2.00 cm ²	45.00 cm

[6] TRACCIÓN PURA

$$\alpha = \beta_1 C \quad c = 0.0 \text{ cm} \quad E_s = 2000000 \text{ kg/cm}^2 \quad \epsilon_c = 0.003 \quad F'_y = 4200 \text{ kg/cm}^2$$

$$P_n = -A_{st} \cdot F'_y = -16 \text{ cm}^2 \times 4200 \text{ kg/cm}^2 = -67200 \text{ kg}$$

$$P_n = -67.20 \text{ tn·m}$$

$$M_n = 0.00 \text{ Ton}$$

[7] RESUMEN DE CARGAS

Ptos.	Mn (tn·m)	Pn (ton)
1	0.00 tn·m	414.83 Ton
2	21.40 tn·m	326.69 Ton
3	25.62 tn·m	277.33 Ton
4	28.02 tn·m	154.30 Ton
5	14.42 tn·m	42.55 Ton
6	0.00 tn·m	-67.20 Ton

CARGAS	
MD =	Ton·m
ML =	Ton·m
Ms =	Ton·m
PD =	Ton
PL =	Ton
PS =	Ton·m

$$1.4 D + 1.7 L$$

$$1.25 (D + L) \pm S$$

$$0.9 D \pm S$$

$$\phi = 0.70$$

$$\alpha = 0.80$$

$$\phi(P_n) = 290.38$$

Pto	Mn (tn·m)	Pn (ton)	$\Phi M_n = M_u$ (tn·m)	$\Phi P_n = P_u$ (ton)	Combinación	Mu (tn·m)	Pu (ton)
1	0.00 tn·m	414.83 Ton	0.0000 tn·m	290.380 Ton	1.4D+1.7L		
2	21.40 tn·m	326.69 Ton	14.9801 tn·m	228.685 Ton	1.25(D+L)+S		
3	25.62 tn·m	277.33 Ton	17.9341 tn·m	194.131 Ton	1.25(D+L)-S		
4	28.02 tn·m	154.30 Ton	19.6137 tn·m	108.010 Ton	0.80D+S		
5	14.42 tn·m	42.55 Ton	10.0926 tn·m	29.787 Ton	0.90D-S		
6	0.00 tn·m	-67.20 Ton	0.0000 tn·m	-47.040 Ton	αP_n		
1'	-14.42 tn·m	42.55 Ton	-10.0926 tn·m	29.787 Ton	$\alpha \Phi P_n$		
2'	-28.02 tn·m	154.30 Ton	-19.6137 tn·m	108.010 Ton			
3'	-25.62 tn·m	277.33 Ton	-17.9341 tn·m	194.131 Ton			
4'	-21.40 tn·m	326.69 Ton	-14.9801 tn·m	228.685 Ton			
5'	0.00 tn·m	414.83 Ton	0.0000 tn·m	290.380 Ton			



Debido a que la sección de la columna circular es uniforme en sentido en X como en Y, se tiene conocimiento de que el diagrama de interacción será igual en ambos sentidos.